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# RADIAL VELOCITIES OF STARS WITH MICROWAVE MASER EMISSION. I. LONG-PERIOD VARIABLES

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## ABSTRACT

Extensive radial-velocity measurements of high-dispersion optical spectra (λλ3500–9000) of long-period variables are presented and analyzed so as to establish the photospheric, emission, and circumstellar velocities. The data are compared with microwave emission velocities from the molecules OH, $H_2O$, and SiO. Correlations among the various velocities are presented and discussed. Usually at least one mircowave component has the same velocity as the stellar photosphere, and frequently one or more microwave velocities agree with the velocity of the circumstellar envelope. Other microwave components frequently fall between the photospheric and circumstellar velocities. The question of whether the absorption or emission lines represent the true stellar velocity is carefully discussed. The correlations among the microwave and stellar absorption velocities favor the absorption velocity as representative of the center of mass. However, the possibility remains that the velocities of the permitted and forbidden lines represent the stellar velocity, provided that the emission lines are formed at small optical depth and hence are strongly limb-brightened.

*Subject headings*: circumstellar shells — long-period variables — radial velocities

## I. INTRODUCTION

The long-period variables have for many years provided an intriguing problem for the construction of a model that can successfully predict the emergent spectrum. The problem became more interesting with the discovery of microwave emission of OH, $H_2O$, and most recently of SiO in many long-period variables, as well as the discovery of an infrared excess in such stars. As an initial step toward understanding the physical processes in long-period variables, it is essential to understand the kinematics, i.e., to decide the depths and motions of regions in which each spectral feature is formed. Only after the kinematics are clear can the dynamics of the atmosphere, the physical processes of excitation—whether they be collisional or radiative—and the method of pumping and masering be understood.

We have therefore undertaken a program for the measurement of radial velocities[1] of long-period variables and related stars with emphasis on those objects which show microwave line emission. The vast amount and variety of radial-velocity data that can be assembled from high-dispersion spectra over a few years requires great caution in presentation, lest a special model or point of view be favored simply by the method of display. We wish particularly to avoid an *a priori* model for which we employ the data to establish numerical values of certain parameters. On the other hand, we prefer not to present only an enormous table of Doppler shifts for individual lines, leaving all interpretation to the reader. The best balance can be achieved if we hypothesize a class of models that will allow us to simplify the data by taking meaningful averages, thus reducing the number of data points to be considered.

The presence of emission lines in long-period variables indicates that a temperature inversion or nonthermal population occurs somewhere in the visible atmosphere of the star. The fact that the emission lines of hydrogen are heavily absorbed by overlying material when they first appear indicates that at least some of the emitting matter lies below the absorption region. During the cycle of variation the overlying asorption lines weaken. Combining this with the 10–15 $km^{-1}$ negative displacement of the emission lines leads to the conclusion that the emitting matter is moving outward through the photosphere. The velocity is supersonic with respect to the photosphere, where $T = 2000-3000$ kelvins, and we may be seeing a fairly weak shock wave.

It is possible, however, that the shock wave is never visible. Double absorption lines, such as are commonly seen in the spectra of type II Cepheids during rising light, are not usually seen in the long-period variables.[2] If the material in front of the shock is optically thin in the continuum and the shock-heated gas is also optically thin, then ab-

[1]All velocities in this paper are measured with respect to the Sun.

[2]There are a few observations of double lines in long period variables. Examples are the S stars R And (Merrill and Greenstein 1958) and χ Cygni (Maehara 1968) and the carbon star R Lep (Phillips and Freedman 1969).



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sorption lines from behind the shock should be visible. If the material in front of the shock is optically thick, the emission lines would be suppressed, while an optically thick shocked gas would emit a hot continuum. The emission lines are not suppressed by overlying continuous absorption (only by line absorption), and no hot continuum is observed. If this line of reasoning is correct, the emission lines may come from the region in front of the shock that is preheated by radiation from the shock wave itself (Whitney and Skalafuris 1963).

There are several other reasons to suspect that some emission lines originate ahead of the shock. The first is the requirement that the kinetic temperature in the region that emits the selectively excited Fe I lines be less than about 3500 kelvins (Willson 1972). The second is the presence of multiplets (2) and (3) of Fe I and the absence of multiplet (41) in emission. No selective excitation mechanism for multiplets (2) and (3) has been proposed or observed. By a comparison of λ4375.93 (multiplet [2], present) and λ4383.55 (multiplet [41], absent), we find a kinetic temperature less than 4000 kelvins, if the excitation of λ4375.93 is thermal. Furthermore, the rather large velocity difference between the Mg II line necessary in order to provide the wavelength coincidence for pumping of λλ4202 and 4307 emission, *viz*, 55 km $s^{-1}$, and the similar velocities required for certain other pumped lines (Willson 1975) indicates a large velocity difference across a shock. Such a high-velocity shock would lead to a high continuum temperature; and emission lines of helium, as seen in W Vir (Wallerstein 1959), would be expected but are not seen. If the Fe I fluorescent lines originate in front of the shock, it is possible that the H I lines are formed in the same region, possibly by absorption of Lyman line radiation. We present these considerations, not as an interpretation of the observed phenomena, but rather as a caveat against the uniqueness of some other interpretations.

We should also hypothesize the possibility of seeing a very extended outer region of the star analogous to the circumstellar envelopes that appear to be pervasive among the M supergiants and early M giants (Deutsch 1960; Weymann 1963). The presence of a moderate temperature inversion as in the chromosphere or possibly a hot corona not directly associated with a shockwave should also be possible, and its location with respect to a cool envelope is not *a priori* obvious. In such a class of models the origin of the microwave emission is not obvious, and it is one of the purposes of this study to elucidate that question.

## II. OBSERVATIONS

Soon after several OH and $H_2O$ sources had been identified with cool stars, observations at high dispersion of such stars and some comparison objects were commenced. Spectrograms were obtained at Hale Observatories, Kitt Peak National Observatory, Dominion Astrophysical Observatory, Mauna Kea Observatory, and Cerro Tololo Inter-American Observatory. All spectra were recorded with conventional photographic techniques except for one series taken at KPNO with the camera of 12-foot (366 cm) focal length and an image tube with extended S-20 sensitivity, yielding a dispersion of 4 Å $mm^{-1}$ at the infrared potassium resonance lines. For a few stars, spectrograms taken by others were found in observatory files and borrowed for measurement.

All measurements were made by the author with oscilliscope measuring engines at the University of California, Berkeley; California Institute of Technology; and the University of Hawaii. Thus no personal equation between measures has entered into the data. Furthermore, by measuring all the stars it was possible to become familiar with their spectra and to recognize anomalies rather readily.

To evaluate the accuracy of the resulting velocities it is essential to understand the statistical errors of measurement and the selection of lines to represent various regions of the stellar envelope discussed in the previous section, and to compare measurements for at least one star with a known circumstellar envelope.

*Statistical Errors.*—For a 20 μ slit at a dispersion of 10 Å $mm^{-1}$ at 4000 Å, the wavelength of a single isolated line should be measureable to about one-tenth of its width, i.e., to ± 1.5 km $s^{-1}$. Blending will undoubtedly increase the errors of measurement. For metallic absorption and emission lines we find standard errors ranging from 1.8 to 3.5 $kms^{-1}$ when we measure a group of similar lines which would be expected to have the same velocity. For hydrogen emission lines the uncertainty is larger due to the breadth of the lines and their disturbance by overlying absorption, and values between 3 and 6 $kms^{-1}$ for the standard errors are common. The accuracy of measurement for the hydrogen lines improves as the star's light diminishes because the overlying absorption decreases and more lines become visible.

Rather than list the probable error of the mean for each measurement in the tables, it suffices to record the following general statements: For hydrogen lines, of which we measure between five and 10 on each plate, the probable error of the mean ranges from 0.6 to 2 km $s^{-1}$. For sharp emission lines, the number measured is always two for silicon and ranges from two to 10 for other elements or ions such as Fe II, implying probable errors of 0.4 to 2.5 km $s^{-1}$. For weak absorption lines it was usually possible to measure at least 10 lines in each group, yielding probable errors of 0.4 to 0.8 km $s^{-1}$. These numbers are for 10 Å $mm^{-1}$ plates and can be modified appropriately for other

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dispersions. For cores of strong lines and isolated lines such as λ6572 of Ca I or λ6707 of Li I, the only empirical indication of accuracy is the consistency from one spectrogram to another. Systematic errors among coudé spectrographs should be minor considering their stable configuration. Velocities based upon one line or on several lines showing unusual scatter are shown in parentheses in the tables.

*Selection and Grouping of Lines.*—A compromise must be reached between listing the velocity of every line on each spectrogram and taking means of so many features as to obscure significant differences. Among the emission lines we have grouped lines according to element, ion, or molecule (only AlH). Velocities for Fe I have generally not been listed in the tables because several lines are selectively excited, and the low-lying transitions, which are probably thermally excited, are clearly disturbed by absorption. Hence their measured velocities are not necessarily indicative of the velocity of the region in which they are formed. Emission at Hα is usually disturbed by an unidentified absorption band, but is included in the tables. The band head appears first at spectral type M6 and becomes sufficiently strong to obliterate the long-wavelength half of the line by type M8. For most long-period variables Hα is not seen, but it is usually present in U Ori, and is seen in other stars, especially when they are going through a maximum that is unusually bright (e.g., R Cas in 1973).

A very large number of absorption lines are present in spectra of long-period variables. For the weak lines we have generally taken separate means of those arising from excited multiplets and those from ground multiplets. The weak absorption lines from excited states are the most likely to represent the ordinary photosphere. Weak ground multiplet lines may be disturbed by circumstellar absorption and hence are recorded separately. The best lines in the blue-violet region for recording circumstellar cores are the strong resonance lines of Cr I and Mn I, when the stars are sufficiently bright for heavily exposed spectrograms of high dispersion to be obtained. In the stars of latest type the Cr I lines are somewhat disturbed by TiO, while the Mn I lines are increasingly affected by violet-displaced emission after maximum light.

In the visual-red region, very few absorption lines are seen for stars later than type M5, which includes most of the spectra on this program. Near type M6 it is still possible to measure the sodium D lines on heavily exposed spectrograms despite some TiO absorption. A conspicuous absorption feature in all M stars is λ6572, the intercombination line of Ca I.[3] Since it arises from the ground state, it may be formed either in the photosphere or the circumstellar envelope. A comparison of the velocities in the tables shows that both situations arise. In R Cas the line appears to be partially resolved into two components on two spectrograms. The resonance line of Li I is present in a few stars. Its wavelength was taken as 6707.822, which corresponds to a partially saturated doublet of $^7$Li for which the stronger line lies at λ6707.761 and the weaker line falls at λ6707.912. The Li I line may be circumstellar, but in R Cas its strength varies with the spectral type of the star, and during the bright maximum of 1974, when the spectrum was unusually early, the line disappeared. Thus it is probably a stellar feature. In the near-infrared the potassium resonance lines are conspicuous. They are frequently circumstellar, or contain a circumstellar component. Also seen in the λλ8100–8800 region are absorption lines of neutral sodium, titanium, and iron. The infrared calcium triplet is weak or absent in the spectra of long-period variables, but is present in supergiants.

*Comparisons with Other Observations.*—As a control, radial-velocity data for α Her A, type M5 II, are shown in Table 1, and may be compared with Deutsch's (1956) data. Deutsch found the following velocities:

Reversing layer: −35.2 km s$^{-1}$
Circumstellar Ca II: −40.8 km s$^{-1}$
Other circumstellar lines: −45.0 km s$^{-1}$.

The difference between Ca II and the other circumstellar lines is discussed by Deutsch and may be due to blending with interstellar lines. Our mean velocity of lines for excited levels differs from his reversing-layer velocity by 0.3 km s$^{-1}$. Our K I, Cr I, and Mn I velocities fall within 1 km s$^{-1}$ of Deutsch's circumstellar velocity, and our Ca II measurement differs by only 1.2 km s$^{-1}$. It is interesting to note that the K I resonance lines are unresolved and yet yield the circumstellar velocity. Evidently the stellar component is dominated by the circumstellar feature; this will appear frequently in the long-period variables. For λ6572 of Ca I, however, the measured velocities are close to the photospheric value. We must be careful in attributing it to a particular layer in the long-period variables.

*Projection Effects.*—For a Cepheid or similar star in which the photosphere is expanding or contracting relative to the center of mass, the measured velocity is less than the true velocity of motion due to the fact that material at the limb is moving perpendicular to the line of sight. When limb darkening is also taken into account, the multiplicative factor that is usually used is 24/17 to yield the true velocity with respect to the center of mass of the star. For an envelope that is much larger than the star, the factor is unity for absorp-

[3]This line is seen in emission in the irregular variable VY CMa, but we have never seen it in emission or even slightly filled in or disturbed in any other star.

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TABLE 1

RADIAL VELOCITIES FOR ALPHA HER

| Plate No. | Date | Disp. (A/mm) region | Absorption | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | Exc. Levels | Ground Levels | | | | | |
| | | | | weak I | K I | Ca I | Cr I | Mn I | Ca II |
| Ec 2915 | 29 Apr. '64 | 16 IR | -34.4 | | -44.8 | | | | |
| 2916 | 29 Apr. '64 | | -34.9 | | -43.3 | | | | |
| Pb 11141 | 2 June '69 | 6.7 R | | | | -33.7† | | | |
| DAO 7469 | 17 June '72 | 10 R | | | | -36.6*† | | | |
| 7470 | 17 June '72 | 10 R | | | | -35.2 | | | |
| DAO 7497 | 30 June '72 | 6 B | -35.1 | -36.3 | | {-33.7 -42.9 | -36.8 -44.3 | -37.3‡ -44.7 | -42.0§ |

*Possibly resolved into two components at −31.2 and −38.5.
†λ 6572 only.
‡Only λ 4030 is clearly resolved.
§Core only.

tion lines. Since there are good reasons to believe that the emission lines in long-period variables are photospheric or subphotospheric, a projection factor should probably be used in the interpretation. The value will depend on the limb-darkening (or brightening for emission lines). We tabulate the measured velocities only, and leave any projection effects to the interpretation.

*Selection of Stars.*—This program was started soon after microwave line emission was discovered; hence the first stars to be observed optically were the first to be found by the radio astrono-

TABLE 2

STARS OBSERVED FOR RADIAL VELOCITY

| Star | 1900 α | δ | ℓ | b | Period (days) | Sp Type* | $V_{LSR} - V_0$ |
|---|---|---|---|---|---|---|---|
| RT Cyg | $19^h40^m47^s$ | +48°-32' | 82° | +12° | 190 | M3e † | +17.8 |
| R Aql | 19 1 33 | + 8 - 5 | 42 | + 1 | 284 | M6.5e | +17.8 |
| R Leo | 9 39 35 | +34 -58 | 191 | +50 | 313 | M7e | - 7.5 |
| S CrB | 15 17 19 | +31 -44 | 49 | +57 | 360 | M6.5e | +16.6 |
| U Ori | 5 49 53 | +20 -10 | 189 | - 3 | 372 | M6.5e | -12.9 |
| W Hya | 13 43 23 | -27 -52 | 318 | +33 | 382 | M8e | + 2.3 |
| R Aqr | 23 38 39 | -15 -50 | 60 | -70 | 387 | M7e | - 1.4 |
| R Hya | 13 24 15 | -22 -46 | 314 | +39 | 388 | M6.5e | + 1.8 |
| U Her | 16 21 22 | +19 - 7 | 35 | +40 | 406 | M7e | +18.3 |
| χ Cyg | 19 46 43 | +32 -40 | 69 | + 3 | 407 | S7,2e | +18.3 |
| R Cas | 23 53 20 | +50 -50 | 115 | -11 | 431 | M7e | + 8.0 |
| IK‡ Tau | 3 47 54 | +11 - 6 | 178 | -31 | 470 | M9e | -12.3 |
| R Dor | 4 35 35 | -62 -17 | 273 | -39 | 338± | M8 III | -16.4 |
| R Crt | 10 55 38 | -17 -47 | 269 | +37 | 160± | M8 | - 7.5 |
| RT Vir | 12 57 34 | + 5 -43 | 310 | +68 | 155± | M8 | + 5.3 |
| RX Boo | 14 19 42 | +26 -10 | 34 | +69 | Irr | M7-8 | +13.5 |
| α Her | 17 10 5 | +14 -30 | 35 | +28 | Const | M5 II | +18.9 |
| HD 207076 | 21 41 24 | - 2 -40 | 54 | -38 | Irr ? | M7 | + 9.3 |
| TW Peg | 21 59 27 | +27 -52 | 84 | -22 | 956 | M7-8 | +12.3 |

*For the long period variables types are at maximum light, mostly from Keenan, Garrison, and Deutsch (1974).

†Even at maximum the type can be from M0 to M5.

‡Formerly known as NML Tau.

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mers. Soon thereafter it was necessary to select among the microwave emitters, and the critereon of optical brightness was very important, as was the season of maximum light. Some stars could be observed near maximum only, while a few observations could be extended to near minimum. Mira was purposely excluded because of the extensive observations by Joy (1954), the fact that it was not found to be an OH or $H_2O$ emitter (though it was later found to be a strong SiO source), and the presence of a faint blue companion that may effect the physical conditions in the envelope. Comparison stars of similar spectral type to long-period variables and long-period variables without microwave emission were originally placed on the program, but were frequently found to be microwave emitters during the four years over which these observations were carried out. The selected stars are listed in Table 2, in order of their periods for the long-period variables, followed by the less regular stars in order of right ascension.

The data for individual stars are presented in Appendix A.

## III. DISCUSSION

The results of this paper are summarized in Table 3 and shown in Figures 1 and 2, wherein we plot the optical and microwave velocities for most of the stars listed in the table. In each case the photospheric absorption velocity is used as a reference. All other velocity data in Table 3 are differences from the photospheric in the sense that a positive entry stands for material moving toward the observer from the photosphere. The microwave velocities are taken largely from Snyder and Buhl (1975). In Table 3 we have done our best to simplify the extensive data listed in the Appendix for each star into one characteristic photospheric,

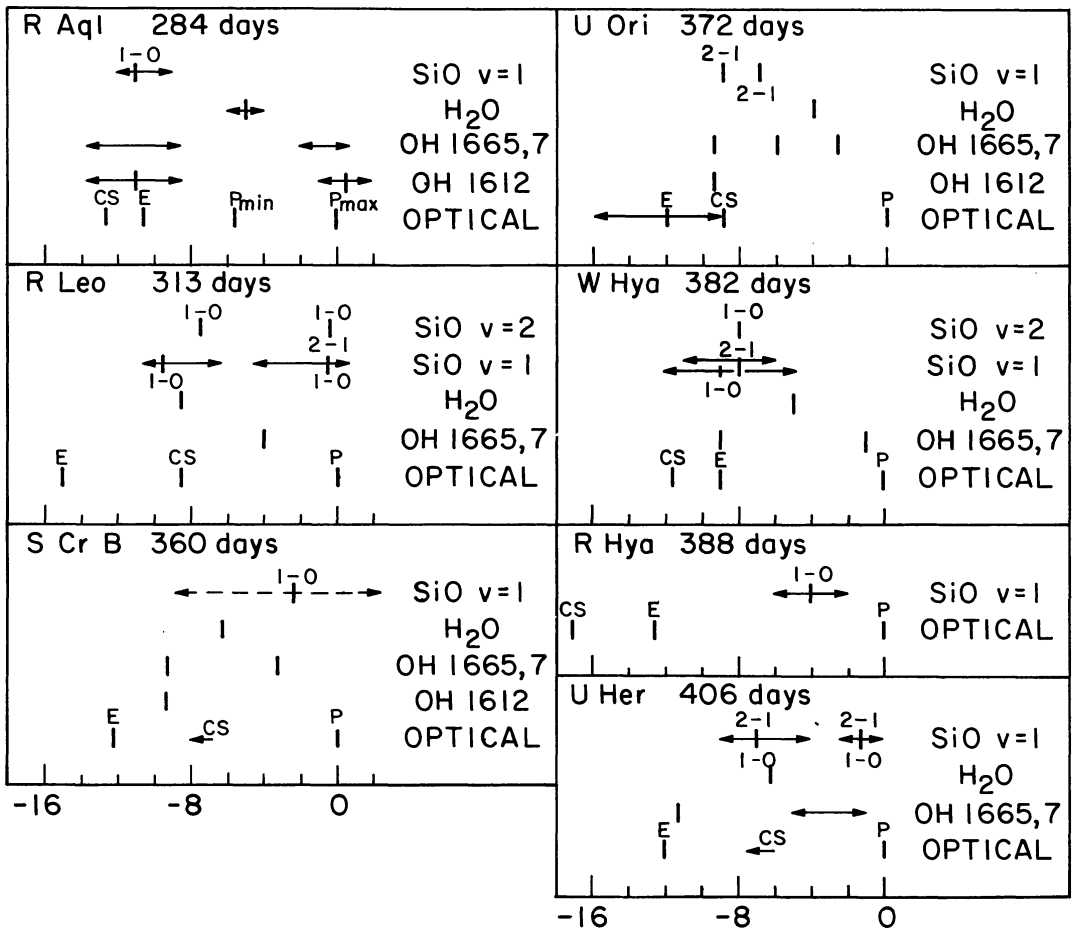


FIG. 1.—A comparison of optical and radio line velocities for seven long period variable stars. For the optical velocities P = photospheric absorption, CS = circumstellar absorption, E = emission. For SiO the vibrational level is shown on the right and the rotational transition is shown above the velocity marker. The arrows indicate the width of noticeably broad features. Uncertain features or velocities are indicated by a question mark or dotted lines. An arrow under CS indicates that the circumstellar velocity is less than indicated but not measured.

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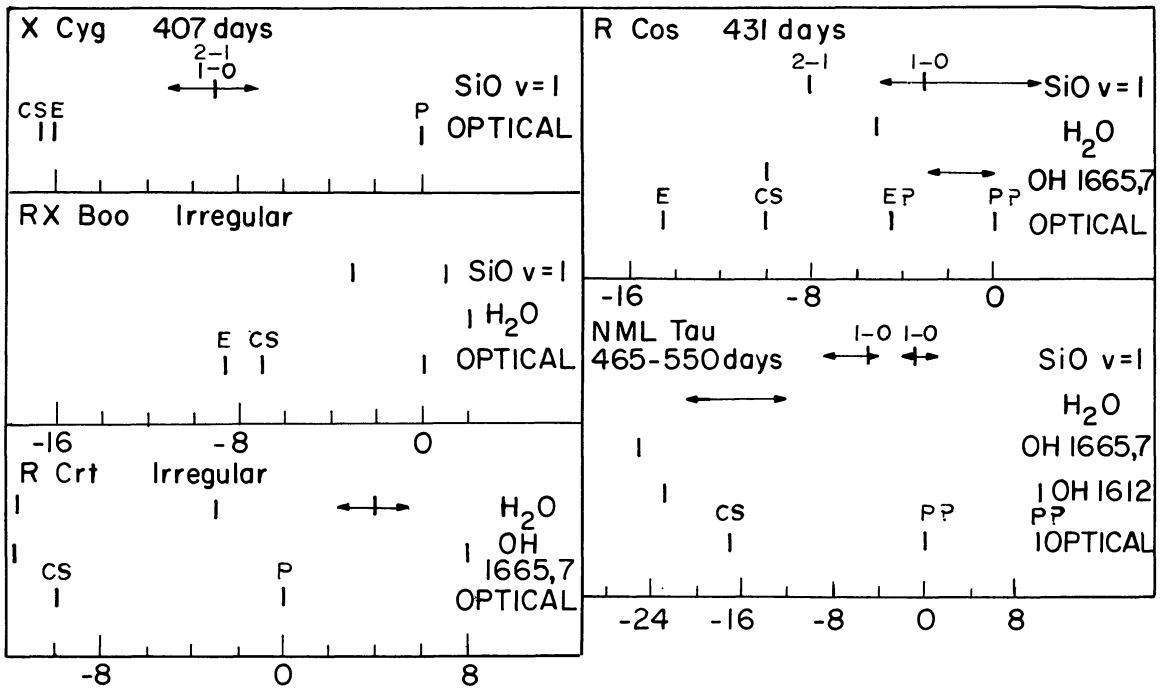


FIG. 2.—Same as Figure 1 for three long period variables and two irregular variables. Note the change of velocity scale for IK Tau.

circumstellar, and emission velocity. For a few stars we have used Merrill's data for assistance. While this process may obscure changes with phase, we need some way to compare the optical and microwave velocities. For no star does the microwave velocity change with phase. While no simple pattern repeats itself throughout the diagrams, a number of correlations can be seen, and some conclusions of interest to theorists may be derived. We state the correlations first and discuss their implications below.

1. For every star except R Crt, TW Peg, and possibly IK Tau, all velocity components are moving outward with respect to the photospheric velocity.

2. Within the limited period range exhibited here, there is little correlation of emission or circumstellar velocity with period. The emission and circumstellar velocities are similar but not identical. The mean difference of (circumstellar − emission) is +1.0 km $s^{-1}$. The observed differences are as large as 6.5 km $s^{-1}$ and are certainly real and not due to errors of measurement. The circumstellar velocities are similar to those found for earlier M giants and supergiants by Deutsch (1960).

3. Where OH at 1612 MHz is seen, the components lie close to the photospheric and circumstellar velocities.

4. For OH at 1665, 1667 the components either lie close to the photospheric and circumstellar velocities or distinctly between the two, except in the case of R Crt, for which one OH component is 8 km $s^{-1}$ more positive than the photospheric velocity.

5. The velocity of $H_2O$ emission usually lies between the photospheric and circumstellar velocities, but there are also examples of $H_2O$ emission at either the photospheric or circumstellar velocity. For R Crt the spread in the $H_2O$ spectrum is unusually large and violates this rule.

6. For the various transitions of SiO, the velocities behave similarly to the $H_2O$ velocities in that they fall either close to or between the photospheric and circumstellar velocity. None of the SiO transitions show a distinct preference for either the photospheric or circumstellar velocity or for the wider available region between the two. R Crt has not been studied in SiO, so we cannot say if it behaves exceptionally in SiO as it does in OH and $H_2O$.

We now discuss the implications of the correlations noted above.

1. The fact that almost all of the velocities are negative with respect to the photospheric velocity indicates that the measured photospheric velocity is indeed the velocity of the center of mass of the star. If the true velocity of the star were to lie within the spread of components shown in Figures 1 and 2, we would expect to find some intermediate objects in which the cause of the shift is only partially operative. If such a shift were due to scattering by moving dust, its effect would be mini-

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TABLE 3

SUMMARY OF VELOCITY DATA FOR LONG-PERIOD VARIABLES AND COMPARISON STARS

| Star | Photo-spheric Velocity (km/sec) | Velocity Differences (km/sec) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | Phot -circum | Phot -em | Phot $-OH_{1612}$ | Phot $-OH_{1665,7}$ | Phot $-H_2O$ | Phot - SiO | | |
| | | | | | | | $v=1$ $J=2\rightarrow1$ | $v=2$ $J=1\rightarrow0$ | $v=1$ $J=1\rightarrow0$ |
| RT Cyg | 115.0 | ? | 10.0* | | | | | | |
| R Aql | 35.5† | 12.5 | 10.5 | -0.5 | 11.5<br>10.5<br>0.5 | 5.5<br>3.5 | | | 12.5 |
| R Leo | 15.0 | 8.5 | 15.0 | | 4.3 | 8.5 | 0.5 | 0.5<br>7.5 | 1.5<br>9.5 |
| S CrB | -9.3 | ≥6.6 | 12.2 | 9.7 | 4.7 | 5.7 | | | 2.2<br>5.3 |
| U Ori | -20.0 | 9.0 | 12.0# | 9.5§ | 9.5<br>2.7<br>6.0§ | 4.0 | 9.0<br>7.0 | | |
| W Hya | +45.0 | 11.5 | ∿9.0 | | 10.5<br>4.0 | 5.0 | 8.0 | 8.0 | 10.0 |
| R Aqr | -15.0 | 13.5 | 13.0 | | | | | | |
| R Hya | -6.5 | 17.0 | 12.5 | | | | | | 4.5 |
| U Her | -27.0 | >6.0 | 12.0 | 0.0"<br>14.0 | (2)<br>(7)<br>(10) | (7) | 1.0<br>7.0 | | 2.0<br>9.0 |
| χ Cyg | 0.0 | 16.5 | 16.0 | | | | 9.0 | | 12.0 |
| R Cas | 23.0 | 10.0 | 14.5 | | 2.0<br>9.0 | 5.0 | 8.0 | | 4.0 |
| IK Tau | 53.0± | 17.0 | | -10.0<br>24.0 | 25.0 | 17.0<br>(16-23) | | | |
| RX Boo | -10.5 | 7.0 | (5-9) | | | -0.5 | 3.0<br>-1.0 | | |
| RT Vir | (13) | (3) | | | | 2.5 | | | |
| TW Peg | {-24.0, -30.5}** | 6.0<br>0.0 | | | | 5 or 11.5†† | | | |
| R Crt | 21.7 | 10.0 | | | 12.0<br>-8.0 | 12.0<br>4.0<br>-3.0 | | | |
| R Dor | 25.0 | 1.5? | 6.7 | | | | | | |
| HD 207076 | -42.0 | 16.0 | (2) | | | | | | |

*From 2 plates of Merrill and one of mine; one plate (approaching min) yields 2 km $s^{-1}$.
†Drops to near 30 at minimum.
±Wing, Spinrad, and Kuhi find 63 ± 5.
§IAU Circ. No. 2680.
"Dickinson, private communication.
#Changes from 16 to 9 from max to min.
**Either variable or one of the spectrograms is in error.
††V = 19 km $s^{-1}$; uncertainty in the difference is due to uncertainty in photospheric velocity.

mized in the infrared, but reference to individual absorption velocities in the tables of data for individual stars does not reveal a systematic difference between the photospheric velocities derived from blue and near-infrared spectrograms. Furthermore, effects of moving dust on the velocity should correlate with the infrared excess; but association of the photospheric velocity with the positive-velocity microwave component in stars with a minimal infrared excess such as R Aql is as common as for extreme stars like VY CMa.

The negative velocities of the microwave lines

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indicate that most of the emitting material is moving outward from the star, and that the material on the far side of the star either is obscured or radiates preferentially in the outward direction and hence is not seen.

The velocities of the forbidden emission lines seen in long-period variables throw doubts upon the conclusion that the absorption lines represent the velocity of the center of mass of the star. If the forbidden lines are formed in an envelope around the star rather than on the surface facing the observer, their velocity should represent that of the entire system, and should not be associated with material that is rising or falling on the near hemisphere.

The commonest forbidden lines seen in long-period variables are those of [Fe II], which strengthen toward minimum light. Near minimum, lines of [S II] and[O I] are also seen (Herbig 1969). To investigate the region of formation for forbidden lines we calculated the volume of space required for the [S II] line at λ4069 to emit 10 percent of the flux in a 0.2 Å bandpass of a model atmosphere with $T_{\rm eff} = 2500$ and $\log g = 1.0$ (Johnson 1974, model J 32). Within the range of conditions limited by $5000 < T_e < 10{,}000$, $10^5 < N_e < 10^7$, H neutral or ionized, we found that [S II] could easily come from the photosphere. The situation for [O I] is less certain. For both λ5577 and λ6300, the emission per unit volume is the same (within a factor of 10), as the flux from [Fe II] and hence [O I] may also be formed in the photosphere, but there are no quantitative data on relative line and continuum intensities for a comparison over the long wavelength interval from λ4300 to λ6300.

There are five reports of velocity data for [S II] and [O I] lines; they are summarized in Table 4. In four out of five cases the forbidden-line velocity is close to the velocity of the other emission lines rather than the absorption lines.

The relationship of both the forbidden and permitted emission lines to the center of mass velocity depends upon the degree of limb-brightening in the lines. The limb-brightening can only be calculated if the depth dependence of the source function is known, which requires a full model of the temperature-reversal region responsible for the emission lines. For sufficiently strong limb-brightening, the emission lines would indeed yield the stellar velocity. Other evidence that the emission velocity may represent the stellar velocity has been presented by Reid (1975).

2. The similarity of emission-line and circumstellar absorption-line velocities indicates that there may be a physical relationship between the two. It would be reasonable for the emitting material to cool as it passes into the circumstellar envelope without much change in its velocity. The extensive emission-line velocity data for U Ori (see Fig. 4) are in agreement with this.

3. The general, but not exact, agreements of the microwave velocities with the optical velocities indicate that the microwave emission may not be coming from a uniform spherically symmetric region. Since the photospheric and circumstellar velocities are derived from absorption lines, they must represent material that covers a substantial fraction of the stellar surface. This requirement may be violated by emission lines, and in fact is likely to be violated if they are masing. Existing data do not justify an attempt to derive a model for such nonsphericity except to express some general considerations. The general agreement of limits on the velocities of microwave lines with photospheric to circumstellar values excludes violent activity such as flares or eruptive prominences. We are probably seeing fairly simple inhomogeneities in the envelope. Density inhomogeneities would greatly affect pumping rates and maser emission rates. Furthermore, the emission may be directional due to maser amplification. The fact that $H_2O$ emission intensity varies while its velocity is constant (Schwartz, Harvey, and Barrett 1974) indicates the lack of motion in the emitting envelope despite continual changes in the pumping rate.

A willingness to accept the possibility of such nonspherical symmetry reduces the need to establish the physical conditions in a variety of layers that are expanding from the star at different velocities.

TABLE 4

VELOCITIES OF [S II] AND [O I] LINES IN LONG-PERIOD VARIABLES
(Velocity in kilometer per second)

| Star | [S II] | [O I] | Absorption | Emission | References |
|---|---|---|---|---|---|
| χ Cyg | -16 | | -5.5 | -14.5 | Merrill (1947) |
| χ Cyg | - 4 | | -3 | -12.5 | Merrill (1953) |
| χ Cyg | | * | | | Herbig (1969) |
| U Her | -37 | | -27 | -39 | This work |
| R Leo | | +5 | +15 | 0 | This work |

*Herbig does not tabulate velocities, but states that the [O I] lines show the same velocity as the emission lines.

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4. SiO emission is surely complex; while good velocity correlations within the various SiO transitions are seen in some stars, there are puzzling differences in others. For example, in R Cas the $v = 1, J = 2 \rightarrow 1$ and $1 \rightarrow 0$ transitions have different velocities. One might expect that molecules making the $J = 2 \rightarrow 1$ transition would immediately make the $1 \rightarrow 0$ transition, yielding a line at the same velocity as seen in $2 \rightarrow 1$. In U Ori two components of $J = 2 \rightarrow 1$ are seen, but no line at $1 \rightarrow 0$ is noticeable. Perhaps a pumping mechanism is present that removes molecules from the $J = 1$ level on a time scale shorter than the lifetime of the level. Since the observations of the $J = 2 \rightarrow 1$ and $1 \rightarrow 0$ transitions were not simultaneous, there may have been intensity or velocity variations between observing sessions. Simultaneous observations of these stars in several SiO transitions over at least one complete cycle may tell us more about the outer envelopes of long-period variables than have any other data so far.

5. The two stars R Crt and TW Peg show at least one microwave emission velocity more positive than the photospheric velocity. Both are semiregular variables with similar spectral types (M7–M8) and weak optical emission lines. They may constitute a recognizable group which should include RX Boo and R Dor. They all are found at high galactic latitudes and would be located very far above the plane unless their luminosities are low. For all of them the Al I resonance lines are strong, perhaps indicating low luminosity for a late M star, and we suggest that they may be less luminous than the long-period variables.

I am deeply indebted to the directors and staffs of the following observatories and institutions for providing observing time and measurement equipment without which this research would have been impossible: Hale Observatories, Kitt Peak National Observatory, Dominion Astrophysical Observatory, Mauna Kea Observatory, Cerro Tololo Inter-American Observatory, Berkeley Astronomy Department, and California Institute of Technology Astronomy Department. This research was supported by several grants from the National Science Foundation, and by the wings of N3341U, which carried me to most of the institutions noted above.

The manuscript was critically read by Dr. W. T. Sullivan, III, and typed (several times) by Mrs. Patricia Hasson Kimpton.

## APPENDIX A

## DETAILED DATA FOR EACH STAR

Since our conclusions depend upon a large number of observed velocities, we present here the measured velocities from each spectrogram. The long-period variables are arranged in order of period and are followed by the irregular variables and comparison stars.

All velocities are measured with respect to the Sun. For corrections to the local standard of rest for each star, see Table 2.

### I. RT CYGNI

This star is a good example of a short-period, high-velocity LPV, and was observed for comparison purposes. Subsequently, weak OH emission was found by Dickinson and Chaisson (1973) but not confirmed by later observations. The velocities on the high-dispersion spectrum DAO 7631 agree with Merrill's (1952*b*) velocities near maximum light, and the less negative emission velocities at phase $+73^d$ show a substantial shift. If the velocity shift of the emission lines toward that of the absorption lines approaching minimum light is confirmed, there are difficulties in understanding the emission lines as due to shock excitation. The emission lines are quite weak, and care should be taken before overinterpreting one spectrogram of limited dispersion.

Observations obtained after the bulk of this research was completed indicate that the velocity of absorption lines in RT Cyg at the maximum in 1974 September correlates with the first ionization potential of the atom. This effect is not seen in any other long-period variable, and may be at least partially responsible for the rather large scatter in the measured absorption lines found by Keenan, Gar-

TABLE A1

RADIAL VELOCITIES FOR RT CYGNI

| Plate No. | Date | Phase (days after max.) | Disp. (A/mm) region | Emission | | | Absorption | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Exc. Levels | Ground Levels | |
| | | | | H | Si I | Fe II | | weak I | Strong |
| DAO 7631 | 13 Sept.'72 | $+33^d$ | 6 B | -125.0 | -127.5 | -121.5 | -116.1 | -115.2 | -117.8 |
| Pd 13113 | 23 Oct. '72 | $+73^d$ | 18 B | -116.0 | -119.0 | -116.2 | -113.6 | -114.6 | -115.0 |

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TABLE A2

RADIAL VELOCITIES FOR R AQUILAE

| Plate No. | Date | Phase (days after max.) | Disp. (A/mm) region | Emission | | | | | Absorption | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | exc. levels | Ground Levels | | | | | | |
| | | | | H | Si I | Fe II | Ti II | Mg I | | weak I | Cr I | Mn I | Ca I | K I | Na I | Ca II |
| D 2372 | 10 Sept.'70 | -62 | 18 R | | | | | | | | | | +31.3 | | | |
| D 2375 | 11 Sept.'70 | -61 | 27 I | | | | | | | | | | | +23.5 | | |
| 2383 | 12 Sept.'70 | -60 | 27 I | | | | | | +35.4 | | | | | | | |
| Ce 20415 | 9 Oct. '70 | -33 | 15 R | | | | | | | | | | +24.7 | | | |
| 20425 | 10 Oct. '70 | -32 | 6 R | | | | | | | | | | +25.6 | | | |
| 20437 | 11 Oct. '70 | -31 | 10 B | +32.9 | +24.0 | | | | +37.7 | +36.6 | +28.9 | +35.1 | | | | |
| Pc 12050 | 10 Dec. '70 | +29 | 4.5 B | +22.9 | +25.7 | | | | | | | | | | | |
| Pb 12056 | 11 Dec. '70 | +30 | 6.7 R | | | | | | | | | | +33.3 | | | |
| Ce 20845 | 13 Oct. '71 | +53 | 10 B | +23.7 | +25.7 | +27.1 | +22.1 | +19.2 | +34.3 | +31.6 | +29.2 | +33.1 | | | | |
| 20855 | 15 Oct. '71 | +55 | 6.6 R | | | | | | | | | | +27.8 | | | |
| Pd 12769 | 26 Feb. '72 | +189 | 18 B | | | | | | +30.4 | +34.5 | +30.0 | +34.5 | | | | |
| Pb 12902 | 26 May '72 | -11 | 6.7 Y | | | | | | | | | | | | +25.3 | |
| Pc 12909 | 28 May '72 | -9 | 13 I | | | | | | | | | | | +21.0 | | |
| 12910 | 28 May '72 | -9 | 9 B | | | | | | +35.6 | +34.4 | +31.6 | +33.0 | | | | |
| DAO 7498 | 30 June'72 | +24 | 6 B | +25.5 | +26.2 | +30.2 | +22.1 | | +34.5 | +33.8 | {+37.6* +26.6 | +36.5* +29.7} | | | | +26.0 |
| 7500 | 1 July'72 | +25 | 10 R | | | | | | | | | | +33.0 | | | |
| 7501 | 1 July'72 | +25 | 10 R | | | | | | | | | | +33.4 | | | |
| Pd 13131 | 26 Oct.'72 | +142 | 18 B | (+29.7) | -- | (+14.0) | -- | +18.8 | +27.8 | +30.5 | +30.0 | +30.0 | | | | |
| D 3708 | 24 May '73 | +75[d] | 4 I | | | | | | | | | | | {+36.7* +23.7 | | |

*Two components, resolved

rison, and Deutsch (1974). Our optical velocities are shown in Table A1.

## II. R AQUILAE

R Aql is one of the shortest long-period variables to show microwave emission. However, its period has decreased steadily from 346 to 282 days in historic times, hence it is probably in some sort of transition state. The amplitude of variations in the infrared ($2-10\mu$) is particularly small (Harvey, *et al* 1974). The emission spectrum is never rich. By 142 days after maximum it is weak, and a spectrum obtained 189 days after maximum showed no emission lines at all. The microwave spectrum is complex and shows such a spread of components that coincidences with optical velocities may well occur due to chance. Our optical velocities are shown in Table A2.

The key to understanding R Aql is the resolution of the K I resonance lines into two components at +37 and +23 km s$^{-1}$ on Kitt Peak spectrogram D 3708. The former velocity agrees with the photospheric velocities and one component of OH (1665, 1667) and the $v = 1, J = 1 \rightarrow 0$ SiO line, while the latter component agrees with partially resolved cores in Ca II, Na I, and Cr I on spectrograms Pb 12902 and DAO 7498.

R Aql is one of the stars for which we have fairly good phase coverage, so we have plotted the velocities as a function of phase in Figure 3. Merrill's (1945) data are included for illustrative purposes, though his velocities appear to be more negative than ours by about 2 km s$^{-1}$. Velocities labeled Mg I refer to the semiforbidden line, λ4571. It is the only emission line measured with reasonable accuracy at phase +142, but it showed a negative residual from the H and Si I lines at phase +53 days, so its use at +142 days as indicative of a velocity change may be questioned.

## III. R LEONIS

We observed R Leo as a comparison star, but it was later found to be a weak OH source at 1665 and 1667 MHz (Fillet, Foy, and Gheudin 1973) as well as an SiO source. Two red spectra and one heavy exposure at 4.5 Amm$^{-1}$ in the blue were obtained; the results are rather ambiguous, but should nonetheless be made available. The latter is heavily exposed and was measured for absorption lines only. The emission lines have been studied in great detail by Merrill (1946*a*, 1952*a*). On the spectrum taken at minimum, a weak emission line is seen at λ6300. If it is [O I] the radial velocity is +4.8 km s$^{-1}$, which is similar to other emission lines. Our optical velocities for R Leo are shown in Table A3.

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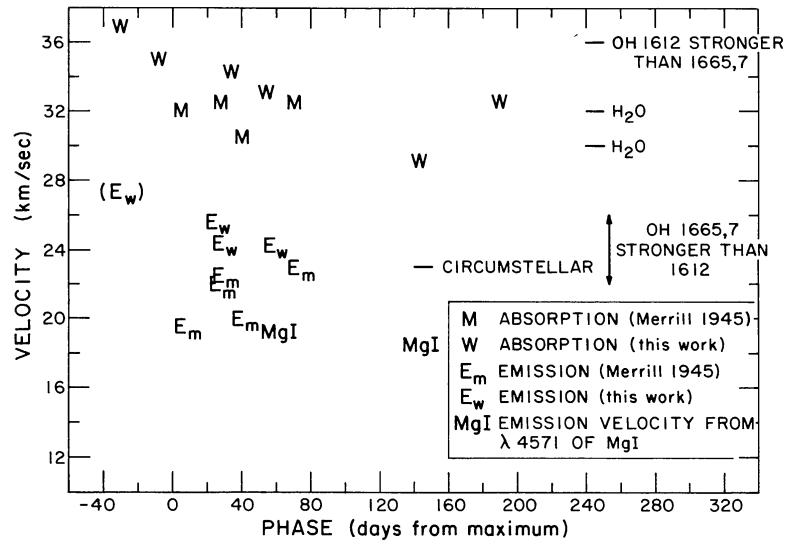


FIG. 3.—The velocity of absorption and emission lines as a function of phase for R Aql. The velocities of OH and $H_2O$ microwave emission lines are shown at the right.

The OH velocity is +10.7 km $s^{-1}$. This falls between the velocity of the absorption lines from excited levels, and the possibly circumstellar velocities as shown by the cores of Cr I and the Ca I line at λ6572. Merrill's absorption velocities, which do not differentiate among lines according to excitation, nor show resolved cores, lie in the range of +11 to +15 km $s^{-1}$. Merrill's velocities are utilized in the discussion and Table 3 of the main paper.

### IV. S CORONAE BOREALIS

This LPV has an inconvenient period of almost exactly a year, but with maximum occurring when it is accessible to observation. Hence spectra, mostly in the blue region, were obtained in 1971 and 1972 near and after maximum light. Microwave emission at the 1612, 1665, and 1667 MHz lines as well as $H_2O$ and SiO are found in this star. The optical velocities are shown in Table A4.

The photospheric absorption velocity, as shown by excited lines, is very nearly −9.3 km $s^{-1}$ from five spectrograms. There is no corresponding sharp microwave feature. An extended envelope is indicated by the fact that even weak zero-volt lines are displaced by 1.8 km $s^{-1}$ to violet, and strong lines show displacements or cores with velocities in the range of −11.0 to −16.6 km $s^{-1}$. These encompass one of the OH 1665, 1667 lines, the $H_2O$ emission, and the maximum SiO emission velocities, but not the OH velocity of −19.0 which is seen in 1665, 1667 and 1612. Unfortunately the only spectrogram in the near-infrared was not sufficiently exposed to show the K I or Rb I lines. Since those lines often show a larger displacement than do the strong blue zero-volt lines or Ca I λ 6572; perhaps they would agree with the −19 km $s^{-1}$ 1612 MHz OH line.

The emission lines in S CrB show somewhat

TABLE A3

RADIAL VELOCITIES FOR R LEONIS

| Plate No. | Date | Phase (days after max.) | Disp. (A/mm) region | Emission [OI] | Absorption: exc. Levels | Ground Levels: weak I | Ca I | Cr I | Mn I |
|---|---|---|---|---|---|---|---|---|---|
| Pc 12053 | 10 Dec. '70 | +186* | 13.5 R | +4.8 | | | +7.3 | | |
| Ce 20639 | 23 Mar. '71 | -21 | 6 R | | | | +1.6 | | |
| Pb 12758 | 25 Feb. '72 | +4 | 4.5 B | | 15.1 | 12.8 | | {6.4, 13.0}† | {10.5, 13.2}† |

*7 days after light minimum.

†cores are listed above, wings below.

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TABLE A4

RADIAL VELOCITIES FOR S CORONAE BOREALIS

| Plate No. | Date | Phase (days after Max.) | Disp. (A/mm) region | Emission | | | | | Absorption | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | exc. levels | Ground Levels | | | | |
| | | | | H | Si I | Fe II | Ti II | Mg I | | weak I | Cr I | Mn I | Al I | Ca I |
| Ce 20642 | 23 Mar'71 | +36 | 10.2 B | {-24.7‡ -16.4§ | -18.7 | -12.0 | | | -9.0 | -11.3 | -13.7 | -11.3 | -14.7‖ | |
| Ce 20645 | 24 Mar'71 | +37 | 15.3 R | | | | | | | | | | | -14.7 |
| Ce 20649 | 25 Mar'71 | +38 | 20.2 IR | | | | | | -9.4 | | | | | |
| Ce 20691 | 15 May'71 | +89 | 10.2 B | -23.8 | -21.7 | -21.0 | | -32.6 | | | | | | |
| Pc 12760 | 25 Feb'72 | +14 | 8.9 B | -20.7 | -24.3 | (-6.3) | | | -9.2 | -8.6 | -11.0 | -11.5 | -11.1 | |
| Pb 12781 | 28 Feb'72 | +17 | 4.5 B | -19.0 | -26.4 | (-7.6) | | | -9.3 | -10.7 | -12.2 | -12.2 | -13.3* | |
| Pc 12904 | 27 May'72 | +106 | 8.9 B | -20.3 | -20.1 | -20.4 | -21.2 | -25.6 | -9.5 | -11.3 | -16.6† | | | |

*One line only.
†Cores only, wings yields velocity −10.5 km s$^{-1}$.
‡Base of lines.
§Point of highest intensity.
‖Cores only, wings yield velocity −11.7 km s$^{-1}$.

greater velocity displacements than do the absorption cores or microwave lines. The hydrogen lines and Si I are near −22 km s$^{-1}$, and the Fe II and Ti II lines are near −21 kms$^{-1}$ after they have clearly emerged. The Mg I line at λ4571 shows a somewhat larger displacement, with velocities of −25.6 and −32.6 on two plates. If these emission lines indicate a flow velocity, there must be a deceleration before the density of the gas reaches a sufficiently low value for OH, $H_2O$, and SiO pumping to be effective.

## V. U ORIONIS

This rather bright variable has the unfortunate period of 372 days, making observations from minimum through rising light impossible in recent years, and observations near maximum are very difficult. Coverage from +60 to +220 days was achieved and improved by measurement of three spectrograms found in the Hale Observatory files. In addition, a Mount Wilson 10.3 Åmm$^{-1}$ spectrogram was kindly obtained by Dr. J. G. Cohen 15 days after the unusually bright maximum of 1973. Our optical velocities for U Ori are shown in Table A5.

The phase coverage is improved if the observations by Merrill (1941, 1945) are included. On high-dispersion plates we have selected the lines of H I and Si I as the easiest to combine with Merrill's measurements for the emission velocity. For Merrill's low dispersion we have used his quoted velocities directly. The velocities are plotted in Figure 4. The emission velocities increase throughout the observable cycle, as in R Leo, reaching the circumstellar velocity, which is also the velocity of the negative components of OH and SiO. It is interesting to note that near minimum, U Ori shows A1H emission and does not show [Fe II]. Usually A1H is seen in S stars but not in stars of type M.

The best evidence for a circumstellar envelope is given by the velocities of the potassium resonance lines, for which a mean of −29 km s$^{-1}$ is found, in agreement with the emission velocity near minimum light and the more negative microwave components. The resonance lines of Cr I and Mn I do not yield evidence for circumstellar components, but the spectrograms are not sufficiently exposed to be favorable for the discovery of cores. The velocity of the Ca I line at λ6572 agrees with the excited and weak zero-volt lines in the blue region. Together these average very nearly to −20 km s$^{-1}$, with no appreciable change with phase. The more positive OH component and the $H_2O$ emission show velocities of −23 and −24 km s$^{-1}$ respectively, which are close to, but not coincident with, the photospheric velocity.

## VI. W HYDRAE

This star is usually called a semiregular variable, but it shows all the spectroscopic features of a long-period variable. It is extremely bright in the infrared, and is a prominent source of $H_2O$ emission as well as OH at 1665 and 1667, though the satellite lines are absent. SiO is also seen. Our optical velocities for W Hya are shown in Table A6.

The absorption velocity pattern is similar to that

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TABLE A5

RADIAL VELOCITIES FOR U ORIONIS

| Plate No. | Date | Phase (days after max.) | Disp. (A/mm) region | Emission | | | | | | Absorption | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Exc. Levels | Ground Levels | | | | |
| | | | | H | Si I | Fe II | Ti II | Mg I | Al H | | weak I | Cr I | Mn I | Ca I | K I |
| Pd 4202 | 29 Nov. '58 | +201 | 18 B | | | | | -31 | -28 | | | | | | |
| 7628 | 30 Oct. '63 | +145 | 18 B | -34.9 | -34.1 | -32.9 | -35.2 | -30 | | | (-19.5) | | | | |
| 7637 | 31 Oct. '63 | +146 | 18 B | -33.3 | -28.3 | -29.4 | | -21 | | | | | | | |
| Pc 11576 | 28 Dec. '69 | +172 | 13.5 R | -57* | | | | | | | | | | | |
| D 2369 | 9 Sept.'70 | +57 | 18 R | -68* | | | | | | | | | | -23.5 | |
| 2379 | 11 Sept.'70 | +59 | 27 IR | | | | | | | -25.6 | | | | | -32.0 |
| 2388 | 12 Sept.'70 | +60 | 9 B | -28.5 | -29.5 | -25.5 | | | | | | | | | |
| Ce 20421 | 9 Oct. '70 | +87 | 6 R | -60* | | | | | | | | | | -22.2 | |
| 20440 | 11 Oct. '70 | +89 | 10 B | -35.3 | -33.3 | -31.4 | -37.2 | -38 | | -21.9 | -21.7 | -24.6 | | | |
| 20450 | 12 Oct. '70 | +90 | 20 IR | | | | | | | | | | | | -30.3 |
| Pc 12052 | 10 Dec. '70 | +149 | 13.5 R | -50* | | | | | | | | | | -24.0 | |
| 12060 | 11 Dec. '70 | +150 | 13.5 IR | | | | | | | | | | | | -25.0 |
| Ce 20854 | 14 Oct. '71 | +84 | 10 B | -33.4 | -30.4 | -28.5 | | | | (-19.1) | (-23.5) | | -21.0 | | |
| 20866 | 23 Nov. '71 | +124 | 10 B | -31.0 | -27.8 | -25.9 | -32.0 | | | | | | | | |
| 20876 | 24 Nov. '71 | +125 | 6 R | | | | | | | | | | | -20.3 | |
| Pc 12756 | 25 Feb. '72 | +218 | 18 B | | | | | -28 | -30 | | | | | | |
| 12771 | 27 Feb. '72 | +220 | 18 B | | | | | -29 | -26 | | | | | | |
| DAO 7635 | 13 Sept.'72 | +57 | 10 R | | | | | | | | | | | -17.8 | |
| 7636 | 13 Sept.'72 | +57 | 10 R | | | | | | | | | | | -17.9 | |
| 7643 | 14 Sept.'72 | +58 | 6 B | -27.2 | -33.2 | -23.5 | | | | -18.9 | -19.8 | -26.0 | -22.6 | | |
| Ce 21693 | 10 Aug. '73 | +15 | 10 B | -34.7 | -35.2 | -38.7 | -34.5 | | | -16.6 | -16.3 | -21.8 | -15.6 | | |

*$H_\alpha$, disturbed by absorption.

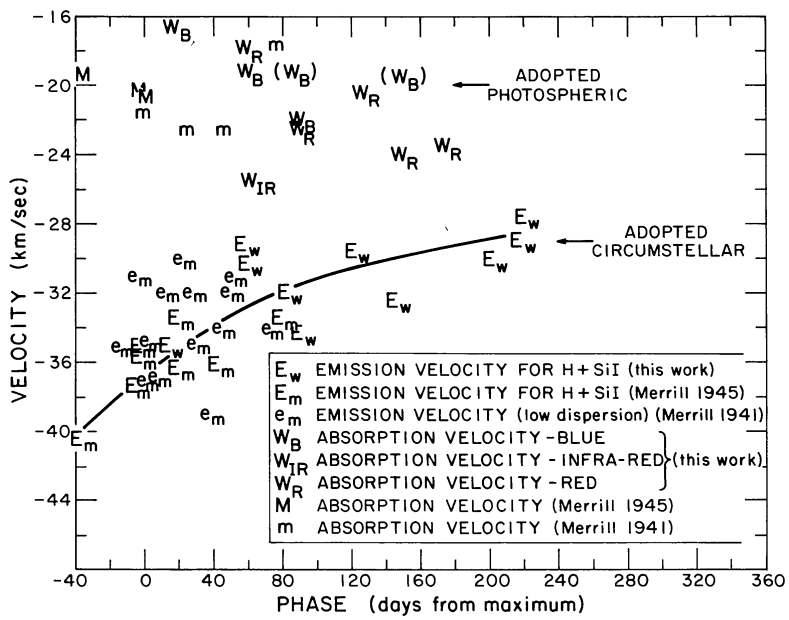


FIG. 4.—The velocity of absorption and emission lines as a function of phase for U Ori. The adopted photospheric and circumstellar velocities (the latter from velocity of line cores not shown on this plot) are indicated.

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TABLE A6

RADIAL VELOCITIES FOR W HYDRAE

| Plate No. | Date | Phase* (days after max.) | Disp. (A/mm) region | Emission | | | | | Absorption | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | Ground levels | | | | | |
| | | | | H | Si I | Fe II | [Fe II] | Mg I | exc. levels | weak I | K I | Mn I | Cr I | Al I | Ca I |
| | 28 Dec'69 | | 13.5 R | | | | | | | | | | | | 36.7 |
| D 31a | 11 Feb'71 | | 17.8 R | | | | | | | | | | | | 37.5 |
| D 34 | 12 Feb'71 | | 8.8 B | 42.9 | 35.3 | (40.7) | | | 43.9 | 42.9 | | 42.4 | | | |
| | 24 Mar'71 | | 20.4 IR | | | | | | | | 36.6# | | | | |
| | 26 Mar'71 | | 10.2 B | {36.8§ 44.6‖ | 37.4 | (43.2) | | | 46.8 | 44.6 | | 42.8 | 39.0 | | |
| Ce 20690 | 16 May'71 | | 10.2 B | 32.9 | 35.5 | 39.2 | | | 44.1 | 44.4 | | 44.4 | 41.4 | {43.0† 54.4‡ | |
| Ce 20694 | 18 May'71 | | 6.7 R | | | | | | | | | | | | 39.5 |
| Ce 20697 | 19 May'71 | | 15.3 IR | | | | | | | | 36.0 | | | | |
| Pb 12774 | Feb'72 | | 4.5 B | 46.5 | 35.4 | (43.1) | | | 44.8 | 44.7 | | {37.9† 43.6‡ | | | |
| KE 725 | Jan'73 | | 6.8 B | | | | 34.6 | 32.9 | | | | | | | |
| D 3698 | 21 May'73 | | 4.0 IR | | | | | | | | 33.8 | | | | |
| 3699 | 21 May'73 | | 4.0 IR | | | | | | | | 33.2 | | | | |

*due to insufficient observations the phases could not be determined.

†partially resolved core.

‡wings.

§velocity as measured near base of lines.

‖velocity as measured at maximum intensity.

#a very weak line at λ7800 probably due to Rb I yields +438 km/sec.

of the low-amplitude semiregulars RX Boo and R Crt. The lines arising from excited levels show a velocity of +44.9 from four plates, while a velocity gradient is evident in that the resonance lines of Mn I show +43.3 while those of Cr I show +40.1. On a 4.5 Å mm$^{-1}$ spectrogram, cores of Mn I are partially resolved at +37.9 km s$^{-1}$. The Ca I line at λ6572 yields +37.9 km s$^{-1}$, and four measurements of the resonance lines of K I show + 34.5 km s$^{-1}$. The photometric and visual data are insufficient to establish the phase of maximum, but the spectra indicate maxima near 1971 February and 1973 March.

Most of the spectra were taken near maximum light, so the emission spectrum is not very well developed and the hydrogen line velocities are disturbed by overlying absorptions, but the Si I emission, which is clear on four spectrograms, shows + 36.3, and Fe II shows a mean of +41.5 (but a value of +39.0 on the only plate with more than one Fe II line). On one plate of 6.8 Å mm$^{-1}$ taken shortly after minimum, the velocity of [Fe II] is +34.6 and of Mg I λ4571 is +32.9. It appears that the emission velocities and circumstellar absorption velocities are very similar.

The microwave velocities span the optical velocity range almost exactly. One component of OH 1665, 1667 and the SiO lines fall close to the emission velocity, with the SiO profile extending to our estimate of the circumstellar velocity. The other OH 1665, 1667 component falls within 1 km s$^{-1}$ of the photospheric velocity. The $H_2O$ velocity clearly falls between the photospheric and circumstellar velocities.

## VII. R AQUARII

The spectrum of R Aqr is peculiar in that nebular lines of [O III] and [Ne III] are seen. In addition, a hot continuum was present at minimum light in the early 1930s (Merrill 1950). Strong polarization was found in the near-ultraviolet during the 1974 minimum (Serkowski 1974). Recent high-dispersion data are shown in Table A7. Additional data and a discussion of the system are to be found in Jacobsen and Wallerstein (1975), wherein it is shown that the emission-line velocities do not yield the spectroscopic orbit suggested by Merrill (1950) . This system may have an envelope whose velocity and excitation increase outward. It does not

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TABLE A7

RADIAL VELOCITIES FOR R AQUARII

| Plate No. | Date | Phase (days after max.) | Disp. (A/mm) region | Emission | | | Absorption | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Ground levels | | | | | | |
| | | | | H | Si I | [ ] | exc. levels | weak I | K I | Ca I | Ca II | Cr I | Mn I | Sr II |
| D 2376 | 11 Sept. '70 | -9 | 27 IR | | | | -16.6 | | -28.6 | | | | | |
| Ce 20417 | 9 Oct. '70 | +19 | 6.7 R | -47.5* | | | | | | -18.8† | | | | |
| Ce 20846 | 12 Oct. '71 | +6 | 10 B | -33.7 | -28.5 | -8.4 | -13.5 | -14.5 | | {-13.5 -26.3 | -48.4Δ | -14.9 | -16.4 | -16.0 |
| Ce 20857 | 15 Oct. '71 | +9 | 6.7 R | -44.0* | | | | | | -17.1† | | | | |
| Ce 20872 | 24 Nov. '71 | +49 | 6.7 R | -39.4* | | | | | | -16.0† | | | | |

*Hα emission may be disturbed by overlying absorption. [ ] = one line of [Ne III] and two lines of [O III]

†λ 6572

Δλ 3968 only.

emit microwave lines, but has been detected in the radio continuum (Gregory and Seaquist 1974).

## VIII. R HYDRAE

R Hya was studied in great detail by Merrill (1946*b*). For that reason, and since it was not at first known to be a microwave emitter, we have obtained only a few spectrograms. SiO emission was discovered later. Our optical emission velocities are similar to those measured by Merrill. The zero-volt absorption lines frequently show displaced, but not resolved, cores, except at the Na D lines, where the cores appear to be fairly clearly resolved near −23.7 km $s^{-1}$. These are not likely to be interstellar, since Münch and Zirin (1961) found interstellar velocities of −2.6 and +18.5 km $s^{-1}$ for the star HD 119608 which is near R Hya. The SiO emission falls between the photospheric and circumstellar velocities. Our optical velocities for R Hya are shown in Table A8.

## IX. U HERCULIS

This rather late-type long-period variable has a period near 406 days and shows emission in the main lines of OH as well as $H_2O$ and SiO. Six spectrograms, two each in the blue, red, and infrared, were obtained.

The photospheric velocity is best established from the 8.9 Å $mm^{-1}$ blue spectrogram at −27.0 km $s^{-1}$ from which the displacement of 2.4 km $s^{-1}$ for weak ground-level lines is probably significant. Cores are displaced, but not resolved, for Cr I and Al I at a velocity near −33 km $s^{-1}$. This is in rather good agreement with both measures of the sharp Li I resonance line and one measure of λ6572 of Ca I, but the other measurement of λ6572 is discrepant. Since the cores are not resolved, it seems safest to conclude that the circumstellar envelope may be moving outward at somewhat more than 6 km $s^{-1}$.

Plate Pd 12900 taken 143 days after maximum, is particularly rich in emission lines; we list their velocities in Table A9. The velocity data for U Her are summarized in Table A10.

TABLE A8

RADIAL VELOCITIES FOR R HYDRAE

| Plate No. | Date | Phase (days after max.) | Disp. (A/mm) region | Emission | | | | Absorption | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | Ground Levels | | | | |
| | | | | H | Si I | Fe II | Mg I | exc. levels | weak I | Cr I | Mn I | Al I | Na I |
| KE -702 | 27 Dec. '72 | +9 | 7 B | -19.1 | -15.4 | -13.9 | -26.5 | -6.5 | -10.0 | {-11.3 -14.0* | -11.4 | {-9.7 -13.3* | |
| KE -731 | 11 Jan. '73 | +24 | 7 Y | | | | | | | | | | {-12.1 -23.7 |
| D -266 | 21 Feb. '73 | +65 | 9 B | -19.2 | -18.4 | -18.1 | -24.5 | -6.5 | -9.9 | -11.2 | {-7.5 -11.4* | | |

*displaced, but not resolved, absorption cores.

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## TABLE A9

EMISSION VELOCITIES FOR U HERCULIS AT +143 DAYS

| Element | Multiplet(s) | No. of lines | velocity (km/sec) | Element | Multiplet(s) | No. of lines | velocity (km/sec) |
|---|---|---|---|---|---|---|---|
| H | | 9 | -39.9 | Fe I | 23 | 2 | -42.2 |
| Mg I | 3 | 3 | 42.5 | Fe I | 24 | 2 | 37.9 |
| Mg I] | 1 | 1 | 41.1 | Fe I | 41 | 4 | 45.9 |
| Si I | 2,3 | 2 | 38.7 | Fe I | 42 | 2 | 42.7 |
| [S II] | 1 F | 1 | 37.1 | Fe I | 68 | 1 | 38.2 |
| Ca II | 1 | 2 | 102.7 | Fe I | 72 & 73 | 8 | 32.4 |
| Sc I | 8 | 1 | 38.5 | Fe I | 388(?) | 1 | 35.3 |
| Ti II | 13,41 | 4 | 40.8 | Fe II | several | 13 | 40.2 |
| Mn I | 2 | 2 | 53.1 | [Fe II] | 7,21 | 4 | 42.8 |
| Fe I | 2,3 | 7 | 46.1 | Co I | 37,58(?) | 2 | 34.9(?) |
| Fe I | 5 | 1 | 54.7 | Ga I | 1 | 1 | 46.2 |
| Fe I | 21 | 2 | -55.0 | In I | 1 | 1 | -33.1 |

## TABLE A10

RADIAL VELOCITIES FOR U HERCULIS

| Plate No. | Date | Phase (days after Max.) | Disp. (A/mm) region | Emission | | | | Absorption | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | exc. levels | Ground Levels | | | | | |
| | | | | H | Si I | Fe II | [Fe II] | | weak I | Cr I | Al I | Mn I | Ca I | Li I |
| Pc 11571 | 28 Dec.'69 | +75 | 13.5 R | | | | | | | | | | -32.0 | -33.0 |
| D 2373 | 11 Sept'70 | -79 | 26.7 IR | | | | | -24.8 | | | | | | |
| Pc 12048 | 8 Dec.'70 | + 9 | 13.5 R* | -35.6 | | | | | | | | | -24.6 | -32.9 |
| Ce 20651 | 25 Mar.'71 | +116 | 20.4 IR | -45.5‡ | | | | | | | | | | |
| Pc 12768 | 26 Feb.'72 | +53 | 8.9 B | -40.7 | -37.5 | -36.4 | | -27.0 | -29.4 | -29.7 } § -32.4 | -24.7 } § -33.7 | -27.8 | | |
| Pd 12900 | 26 May '72 | +143 | 17.8 B | -39.9 | -38.7 | -40.2 | -42.8 | | (-34.5) | | | | | |

*Also 8.9 B at Hδ which yielded the H velocity as shown.
‡Only one line; three Ca II lines yielded −40.0 km s$^{-1}$
§Separate measures refer to wings (*above*) and core (*below*) not resolved as separate entities.

## TABLE A11

RADIAL VELOCITIES FOR χ CYGNI

| Plate No. | Date | Phase (days after Max.) | Disp. (A/mm) region | Emission | | | Absorption | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Exc. Levels | Ground Levels | | | | | | |
| | | | | H | Si I | Fe II | | weak I | Ca I | Cr I | Mn I | K I | Rb I | Na I |
| D 2367 | 9 Sept'70 | -51 | 18 R | | | | | | -7.6 | | | | | |
| D 2382 | 12 Sept'70 | -48 | 27 IR | | | | +6.6 | | | | | -16.7 | -7.3 | |
| Ce 20416 | 9 Oct.'70 | -21 | 6 R | -36* | | | | | -7.4 | | | | | |
| Ce 20865† | 23 Nov.'71 | -31 | 10 B | -16.0 | -19.9 | (-2.9) | +0.5 | -3.1 | -10.5‡ | { -6.7 -15.8 } | -0.2 | | | |
| Ce 20871 | 24 Nov.'71 | -30 | 6 R | | | | | | -5.1 | | | | | |
| Pc 12647 | 16 Dec.'71 | - 8 | 9 B | -14.5 | -13.5 | (+0.6) | -0.5 | -3.1 | { -3.9 -16.9 } | { -5.5 -13.2 } | { -2.7 -9.7 } | | | |
| Pc 12650 | 17 Dec.'71 | - 7 | 13.5Y | -19§ | | -13.5 | -4.1‖ | -4.1‖ | | | | | | { -7.8 -16.0 } |

*Hα, velocity disturbed by absorption.
†Ca II absorption core at −14.3 km s$^{-1}$.
‡λ4226.
§From Hβ only; emission at Hα shows −33 km s$^{-1}$
‖Only two lines measured of each class.

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### X. CHI CYGNI

Chi Cygni is not an OH or $H_2O$ emitter; however, it was observed for comparison purposes and later found to be an SiO maser. The optical data are presented in Table A11. The absorption velocities span the range from near zero for lines arising from excited levels, to near −16 km $s^{-1}$ for cores of strong zero-volt lines. The latter are K I, Cr I, Ca I, Ca II, and Na I. There is probably a velocity gradient in the outer envelope of $\chi$ Cygni. The emission lines are weak and/or absorbed by overlying gas, since all of the spectrograms were obtained before maximum light; however, their velocities are similar to those recorded by Merrill (1947, 1953). It is interesting to note that for $\chi$ Cyg, the emission velocity and circumstellar absorption velocity are nearly the same. The emission velocity remains nearly constant throughout the cycle (except near minimum when it may rise to reach the SiO emission velocity) as contrasted with U Ori (see Fig. 4). Merrill's and our data show this very well, and we illustrate it in Figure 5, wherein velocities of hydrogen, Fe II, [Fe II], and the intercombination line of Mg I at λ 4571 are shown. There is substantial scatter near minimum, where the dispersion is limited, and for λ4571 at all phases. The SiO velocity lies between the photospheric and circumstellar velocities.

Double absorption lines have been seen in $\chi$ Cyg near maximum light by Maehara (1968). Their behavior is similar to that of the double lines in the W Virginis stars because the shortward-displaced components are first weak, and during the cycle from 18 days before to 66 days after maximum, they increase in strength while the longward component weakens. Furthermore, the shortward components show a higher excitation temperature, indicating that they are formed at a deeper level than are the longward components. It is interesting to note that the violet absorption components join the emission velocities near phase +80 days.

### XI. R CASSIOPIAE

The absorption velocities of this star of relatively long period, 431 days, provide good evidence for a circumstellar envelope moving outward at 10 km $s^{-1}$. Resolved cores are seen in Mn I and Cr I lines in the blue and at the Na D lines. Weak zero-volt lines, including the moderately strong Ca I and Li I lines in the red, show velocities near the photospheric velocity, though the Ca I line appears to be partially resolved on two spectrograms. Our optical velocities for R Cas are shown in Table A12.

The optical emission lines in R Cas present a minor quandary. The velocities near 18 km $s^{-1}$ at 155 days after maximum light are seriously ar variance with the velocities at +20 days, two high-dispersion velocities of Merrill (1945), and 12 low-dispersion velocities of Merrill (1941), but we know of no reason to discard or disbelieve them.

The microwave velocities show one OH 1665, 1667 component and a broad SiO, $v = 1, J = 1 \rightarrow 0$ line near the photospheric velocity. The other OH 1665, 1667 component falls nicely on the circumstellar velocity; and the SiO, $v = 1, J = 2 \rightarrow 1$ line is

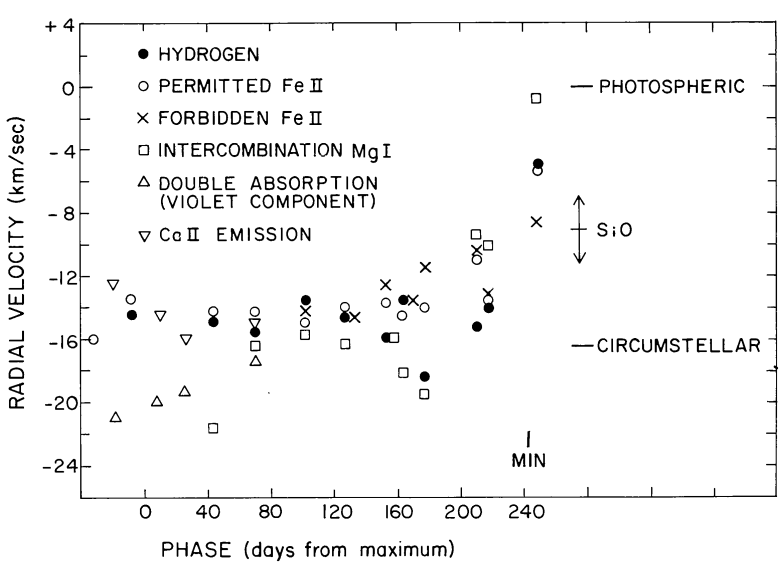


FIG. 5.—The velocity of various emission lines in $\chi$ Cyg. The photospheric velocity, which is constant with phase, circumstellar velocity, and SiO emission velocity are also shown. In addition, Maehara's (1968) velocities for the short absorption components during phases of double lines as well as his velocities for the infrared Ca II lines are shown.

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TABLE A12

RADIAL VELOCITIES FOR R CASSIOPEIAE

| Plate No. | | Date | Phase (days after max.) | Disp. (A/mm) region | Emission | | | | | Absorption | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | H | Si I | Fe I | Fe II +Ti II | Mg I | exc. levels | Ground Levels weak I | Cr I Mn I | Ca I | Li I | Na I |
| DAO | 7502 | 1 July 72 | +41 | 10 R | | | | | | | | | 20.6* | 24.5 | |
| | 7503 | 1 July 72 | +41 | 10 R | | | | | | | | | 18.8† | 21.8 | |
| | 7642 | 14 Sept 72 | +116 | 10 R | | | | | | | | | 18.7 | 22.0 | |
| Pd | 13114 | 23 Oct 72 | +155 | 18 B | 17.1 | 18.8 | 16.3 | 18.8 | 15.9 | (19.5) | 22.3 | 22.3‡ | | | |
| | 13115 | 23 Oct 72 | +155 | 18 B | 19.1 | 19.5 | 17.6 | 18.7 | 13.3 | (16.1) | 22.4 | 21.7‡ | | | |
| DAO | 8456 | 28 July 73 | +20 | 6 B | 7.8 | 7.0 | 5.4 | 8.4 | | 23.4 | 22.2 | 13.5 | | | |
| | 8457 | 28 July 73 | +20 | 6 B | 6.3 | 12.1 | 4.9 | 4.8 | | 23.1 | 22.8 | {23.0 13.1} | | | |
| | 8459 | 29 July 73 | +21 | 10 R | | | | | | | | | 18.7 | | 13.0§ |
| | 8460 | " | +21 | 10 R | # | | | | | | | | 19.2 | | |
| | 8461 | " | +21 | 10 R | | | | | | | | | | | 11.0‖ |
| | 8462a | " | +21 | 10 R | # | | | | | | | | 20.5 | | |
| | b | " | +21 | 10 R | # | | | | | | | | 21.4 | | |
| | c | " | +21 | 10 R | # | | | | | | | | 20.6 | | |

*partially resolved, components at about 17.0 and 25.6 km/sec

†partially resolved, components at about 16.1 and 22.9 km/sec

‡from Cr I only; Mn I lines disturbed by emission

§red component not resolved, center of blend at 18.9 km/sec

‖red component not resolved, center of blend at 14.8 km/sec

#Hα is present with a velocity near -14 km/sec. The velocity is almost certainly falacious since the line is dissected by a molecular absorption band.

within 2 km $s^{-1}$ of the circumstellar velocity. The $H_2O$ velocity falls between the photospheric and circumstellar velocity.

This star is a good example of the lack of association of the optical emission velocity with any microwave feature. Most likely the optical emission velocity lies 14 km $s^{-1}$ to the violet of the photospheric velocity, and no microwave feature is seen within 4 km $s^{-1}$ of the emission velocity. If the emission velocity shows a positive shift of 10 km $s^{-1}$ at other times, as indicated by our measurements at +155 days, then its association with microwave emission is excluded by its very shifting, since microwave lines are observed to change intensity but not to change velocity.

## XII. IK TAU (= NML TAU)

The period of this star has been quoted as ranging from 465 to 550 days; it has the longest period of any long-period variable in this study.

We have one 27 Å $mm^{-1}$ near-infrared spectrogram, D 2378, taken near maximum light in 1970. The results are unfortunately ambiguous, and we list the wavelengths of individual absorption lines, since spectra of this faint star are very hard to obtain. If the last-listed line, λ8387 of Fe I, may be disregarded as a possible blend, we can estimate a photospheric velocity near +53 km $s^{-1}$ from the excited Na I lines and a circumstellar velocity near +36 km $s^{-1}$ from the resonance lines of K I and Rb I. From spectra near 1.0 μ Wing, Spinrad, and Kuhi (1967) found a velocity of +63 ±5 km $s^{-1}$, in fair agreement with our photospheric velocity. For IK Tau, Wilson, Barrett, and Moran (1970) find OH emission at +30 and +63 km $s^{-1}$, in reasonable agreement with our velocities and in excellent agreement with the photospheric velocity of Wing *et al.* (1967). Our optical velocities for IK Tau are shown in Table A13.

## XIII. R DORADUS

This semiregular variable is of very late type, M8, and probably a little cooler than is RX Boo. It

TABLE A13

VELOCITIES OF INDIVIDUAL ABSORPTION LINES IN IK TAURI (=NML TAURI)

| $\lambda_{lab}$ (Å) | IDEN. | lower E.P. (eV) | Vel (km/sec) |
|---|---|---|---|
| 7698 | K I | 0.0 | +33.7 |
| 7800 | Rb I | 0.0 | +38.6 |
| 8183 | Na I | 2.10 | +50.1 |
| 8194 | Na I | 2.10 | +56.4 |
| 8387 | Fe I | 2.17 | +36.1 |

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has not been reported to be an OH or $H_2O$ emitter. The visual amplitude is no more than a magnitude, with a quasiperiod of 338 days, or perhaps 169 days. Photoelectric observations of R Dor for 1969–1971 are available (Eggen 1973), but it was not followed sufficiently for the AAVSO to provide phases for 1972–1973. The spectrum is surprising in that it looks rather like that of a long-period variable near maximum light. Hydrogen emission is present, and it suffers absorption by overlying metals. Other emission lines are due to Si I, Fe II, Fe I (selectively excited), and even In I excited by Hδ emission. From two 9 Å $mm^{-1}$ plates, one of which was heavily exposed, there is only slight evidence of displaced circumstellar absorption cores of Mn I. The Cr I lines are disturbed by TiO. The emission lines are displaced by about 7 km $s^{-1}$ to the violet of the absorption lines, and are absorbed by overlying material. Previous observations showed very similar absorption velocities, while a few emission lines were seen by the Lick Observers on three plates (Campbell and Moore 1928), but none were seen by Feast (1963) on one spectrogram. This star deserves to be monitored for changes in emission-line intensity and for comparison with long-period variables. Our optical velocities for R Dor are shown in Table A14.

### XIV. RX BOOTIS

This M7–8 star is typical of the semiregular or irregular variables of small amplitude whose absorption spectra are virtually indistinguishable from those of the long-period variables. Joy (1942) found it to show strong Si I and weak H emission, but our observations during 1970–1972, consisting of 11 spectrograms, five of which were taken in the blue-violet region, showed weak emission on only two occasions. The optical velocity data for RX Boo indicate a constant photospheric velocity near −10.5 km $s^{-1}$ with cores of zero-volt lines displaced (but not resolved) by varying amounts up to 9 km $s^{-1}$. Giving high weight to the potassium lines, we estimate that the circumstellar envelope is moving outward at about 7 km $s^{-1}$. The emission velocity is difficult to estimate, but probably lies in the range of −15 to −19 km $s^{-1}$, i.e., close to the circumstellar velocity. The weakness of Hβ indicates that the emission is subphotospheric, not chromospheric. Our optical velocities for RX Boo are shown in Table A15.

The microwave spectrum of RX Boo consists of $H_2O$ with a velocity 2 km $s^{-1}$ more positive than the photospheric velocity. One SiO line falls within 1 km $s^{-1}$ of the photospheric velocity, while the second falls between the photospheric and circumstellar velocities. The uncertainties of measurement would permit the photospheric velocity and the velocities of $H_2O$ and one SiO component to be the same.

### XV. RT VIRGINIS

This cool, semiregular variable may be similar to RX Boo and R Dor. It was added to the program after $H_2O$ emission was discovered. From two image-tube spectra we derive a mean velocity for the K I resonance lines of +9.9 km $s^{-1}$, in excellent agreement with the $H_2O$ velocity of +10.4 km $s^{-1}$. From four spectrograms Joy (1942) derived an absorption velocity of +13 km $s^{-1}$; which probably represents the photospheric velocity. Our optical velocities for RT Vir are shown in Table A16.

### XVI. R CRATERIS

This semiregular variable has a spectrum similar to that of RX Boo and R Dor. There were no emission lines present; the velocities are given in Table A17. A velocity gradient appears to be present. The weak lines from excited levels, weak zero-volt lines, and red lines of Ca I and Li I yield a mean velocity of 21.4 km $s^{-1}$. Strong zero-volt lines in the blue region, i.e., Cr I, Mn I, and Al I yield 17.9 km $s^{-1}$, while the one infrared plate showed 12.0 km $s^{-1}$ from one line each of K I and

TABLE A14

RADIAL VELOCITIES FOR R DORADUS

| Plate No. | Date | Disp. (A/mm) region | H | Emission | | | Absorption | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Si I | Fe II | Fe I | exc. levels | Ground Levels weak I | Mn I | Cr I |
| D 164 | 2 Mar. '72 | 9 B | 17.4 | 18.5 | 22.9 | 18.2 | 26.5 | 26.8 | 26.6 | |
| D 264 | 21 Feb. '73 | 9 B | 18.9 | 16.0 | 20.6 | (18.1) | 25.0 | 24.3 | {23.4* 28.1† | (24.3) |

*core

†wings

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TABLE A15

RADIAL VELOCITIES FOR RX BOOTIS

| Plate No. | Date | Disp. (A/mm) region | Emission | | | Absorption | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | exc. levels | Ground Levels | | | | | |
| | | | H | Si I | Fe II | | weak I | Mn I | Cr I | Ca I | Al I | K I |
| Pc 12055 | 9 Dec'70 | 9 B | | | | -10.7 | -10.5 | -13.3 | -12.1 | | -10.8 | |
| Pc 12054 | 9 Dec'70 | 13.5 R | | | | | | | | -16.3* | | |
| Ce 20641 | 22 Mar'71 | 10.2 B | | | | - 9.2 | - 8.5 | - 9.2 | -11.4 | | | |
| Ce 20640 | 22 Mar'71 | 6.7 R | | | | | | | | -15.4* | | |
| Ce 20693 | 17 May'71 | 6.7 R | | | | | | | | -14.2* | | |
| Ce 20696 | 18 May'71 | 10.2 B | -15.9 | (-14.4) | -14.8 | -11.6 | | -12.1 | -12.9 | | | |
| Ce 20698 | 18 May'71 | 15 IR | | | | - 9.0 | | | | | | |
| Pc 12761 | 25 Feb'72 | 9 B | (-24.5) | -19.2 | (-8.9) | - 8.8 | -10.6 | - 9.2 | -15.2† | | -12.7 | |
| Pc 12903 | 27 May'72 | 9 B | | | | -11.2 | -12.2 | -17.1† | -13.4 | | -19.3† | |
| Pc 12907 | 28 May'72 | 13.5IR | | | | -11.6 | | | | | | |
| Pc 12908 | 28 May'72 | 13.5IR | | | | -11.1 | | | | | | -17.3 |
| DAO 7471 | 17 June72 | 10 R | | | | | | | | -12.7* | | |
| D 3703 | 22 May'73 | 4 IR | | | | | | | | | | -17.9 |

*λ 6572

†cores rather well defined.

TABLE A16

RADIAL VELOCITIES OF RT VIRGINIS

| Plate No. | Date | Disp. (A/mm) region | Absorption Lines |
|---|---|---|---|
| | | | K I |
| D 3701 | 21 May '73 | 4 IR | +7.7 |
| 3707 | 24 May '73 | 4 IR | +12.1 |
| | | | mean + 9.9 |

Rb I. The strong zero-volt lines in the blue are blends, rather than resolved cores, and hence are likely to show an intermediate velocity.

The microwave velocities are complex (Dickinson, Bechis, and Barrett 1973; Dickinson, private communication). OH is present at 10 and possibly 30 km $s^{-1}$, while $H_2O$ is variable in intensity, with components seen at one time or another at 18 and 25 $km^{-1}$.

TABLE A17

RADIAL VELOCITIES FOR R CRATERIS

| Plate No. | Date | Disp. (A/mm) region | Absorption | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Exc. levels | Ground levels | | | | | | | |
| | | | | weak I | Mn I | Cr I | Al I | Ca I | Li I | K I | Rb I |
| Pc 12766 | 26 Feb. '72 | 13.5 R | | | | | | 22.8 | 24.7 | | |
| Pc 12773 | 27 Feb. '72 | 13.5 R | | | | | | 18.3 | 19.8 | | |
| Pc 12780 | 28 Feb. '72 | 9 B | 21.4 | 20.5 | 18.2 | 17.5 | 18.6 | | | | |
| KE - 710 | 28 Dec. '72 | 6.8 B | 22.0 | 21.8 | 18.4 | (15.9) | | | | | |
| KE - 720 | 9 Jan. '73 | 13.5 IR | 21.6 | | | | | | | 12.9 | 11.1 |

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The OH and $H_2O$ components displaced by 12 km $s^{-1}$ from the photospheric velocity agree reasonably well with the circumstellar velocity, which is not determined very accurately. There is also an $H_2O$ component with a negative displacement of 3 $kms^{-1}$ from the photospheric velocity, which is a common pattern. However, the $H_2O$ component positively displaced by 5 km $s^{-1}$ and the OH component similarly displaced by 10 km $s^{-1}$ are unusual among the long-period variables and related stars. The positively displaced OH component was observed only once, however, and its reality has been questioned by Dickinson *et al.* (1973). This would be an important star to observe at the SiO lines.

### XVII. HD 207076

This M7 III star was observed as a comparison star since it was one of the latest type nonvariable M giants in the sky, but more recent observations by Eggen (1973) show it to be irregularly variable, with a visual amplitude of 0.3 mag. The spectrum does not appear to be very different from RX Boo or R Crt. Our optical velocities for HD 207076 are shown in Table A18.

The velocities indicate a circumstellar envelope moving outward at about 16.5 km $s^{-1}$ as measured by the K I lines, both of which were measurable on two exposures on plate D 2385. The Ca I line at λ 6572 shows a velocity that falls between the circumstellar and stellar components. The blue spectrogram, Ce 20878, was very heavily exposed, and some displacement of strong resonance lines is noticeable, but cores are not yet resolved. One emission line of Fe II at λ3938 is present and yields a velocity concordant with the absorption lines. A weaker exposure of 1971 May 16 showed no emission lines or cores, and was not measured.

### XVIII. TW PEGASI

This semiregular variable of period 956 days was added to the program after strong $H_2O$ emission was found by Dickinson (private communication). The optical velocity data are given in Table A19, and are difficult to interpret because of the apparent change in velocity of the photospheric lines between 1973 September and December. Joy (1942) reported a velocity of −27 km $s^{-1}$ from

TABLE A18

RADIAL VELOCITIES FOR HD 207076

| Plate No. | Date | Disp. (A/mm) region | Emission Fe II | Absorption | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | exc. levels | Ground Levels | | | | | |
| | | | | | weak I | K I | Ca I | Cr I | Mn I | Sr II |
| D 2385 | 12 Sept. '70 | 27 IR | | -42.4 | | -58.5 | | | | |
| Ce 20426 | 10 Oct. '70 | 6.7 R | | | | | -48.3* | | | |
| Ce 20856 | 15 Oct. '71 | 6.7 R | | | | | -48.2* | | | |
| Ce 20878 | 25 Nov. '71 | 10 B | -44.2 | -41.3 | -42.5 | | -44.1 | -44.1 | -46.5 | -44.6 |

*λ 6572

TABLE A19

RADIAL VELOCITIES FOR TW PEGASI

| Plate No. | Date | Disp. (A/mm) region | Absorption Velocities (km/sec) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | Excited levels | | Ground Levels | | | | | |
| | | | weak | $H_\alpha$ | weak I | K I | Cr I | Mn I | Al I | Ca I |
| D 3702 | 22 May 73 | 4 IR | | | | -31.3 | | | | |
| DAO 8463 | 29 July 73 | 10 R | | -22.0 | | | | | | -26.0 |
| DAO 8738 | 29/30 Sept 73 | 19.8 B | -30.6 | | -29.3 | | -33.2 | -31.5 | -25.2 | |
| DAO 8739 | 30 Sept 73 | 10 R | | -28.5 | | | | | | -31.0 |
| Ce 21786 | 14 Dec 73 | 10 B | -24.1 | | -24.6 | | {-20.7, -29.9}* | -28.3 | | |

*one Cr I line appears to be resolved into two components

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three low-dispersion spectrograms. The velocity near −30 km $s^{-1}$ from DAO 8738 might be questioned on the grounds of its relatively low dispersion, but 20 weak lines were measured, which should be sufficient. H$\alpha$ is clearly present in absorption in 1973 July when the spectral type is about M6, which is rather late for a star with H$\alpha$ absorption. H$\alpha$ is weak by 1973 September and absent on a lightly exposed spectrogram taken in 1973 December. During this time the spectral type became a little later, about M7, and the star faded by over a magnitude. The H velocities are reasonably concordant with the photospheric absorption velocity. Considering the long period, this star may be a supergiant; but its galactic latitude of −22° would give it a distance of over 1 kpc above the galactic plane, for a modulus of about 13 magnitudes.

If the photospheric velocity is near −31 km $s^{-1}$, there is no evidence for an expanding envelope. If the photospheric velocity is variable or close to −25 km $s^{-1}$, an envelope expanding at 6 or more km $s^{-1}$ is probably present. The $H_2O$ feature which was strong near maximum light in the spring of 1973 died out as the star faded, and had not yet reappeared as of 1974 October. Its velocity of −20 km $s^{-1}$ is more positive than any of the optical velocities. Since the cycle is long, this star clearly deserves further observation, but the similarity to R Crt is already notable.

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